

A380X DIRECT TO options

DIRECT WITH ABEAM, CRS IN and CRS OUT on the MFD DIRECT TO page, as in the A380 FCOM

Aircraft	A380X
Where to find it	MFD > ACTIVE > F-PLN > DIR TO (or the KCCU DIR key)
PR branch	pr/14-direct-to-options (stacked on PR 10)
Status	Built with 10 unit tests on the flight plan geometry; checked in the sim during development.
Pictures	Captain MFD in the local browser harness, running the build installed in the sim. The harness has no navigation database and no flight plan, so the geometry is shown with a diagram (figure 3).

1. What it is for
2. Using the page
3. What to expect
4. Limitations and why

1. What it is for

The DIRECT TO page of the A380 (FCOM DSC-22-FMS-20-30, DIRECT TO) has four options. FBW showed them but only the first one worked. All four work now:

Option	What the FMS builds
DIRECT	A direct leg from the aircraft to the target waypoint (as before).
DIRECT WITH ABEAM	The direct leg, plus an abeam point for each waypoint it bypasses: the point of the direct leg abeam that waypoint, named AB + the first 5 letters of the waypoint (BERGI gives ABBERGI).
CRS IN	An inbound course to the target: the aircraft continues on its current track to the intercept point INTCPT on that course, then flies the course to the target.
CRS OUT	An outbound course from the target: the aircraft continues on its current track to the intercept point INTCPT on the outbound course, then follows it (without overflying the target).

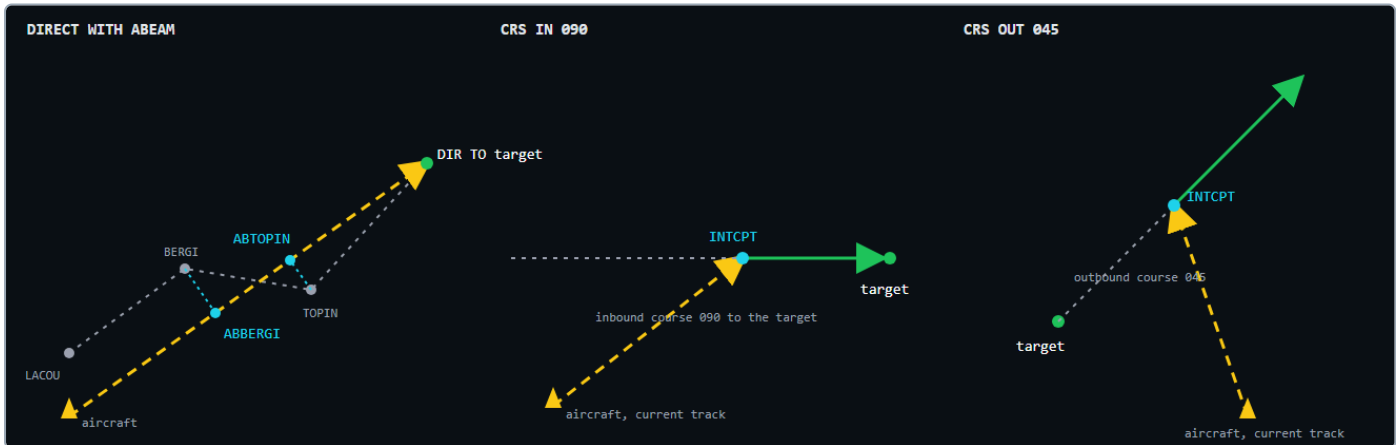


Figure 1. The three new options. Yellow: the aircraft and its path to the first point; green: the course flown after it; blue: the points the FMS creates.

2. Using the page

1. Choose the target in **DIRECT TO** (a flight plan waypoint from the list, or type any fix).
2. Choose the option. For CRS IN / CRS OUT a course field appears next to the option.
3. For a flight plan waypoint the course is already filled: CRS IN = the course from the previous waypoint to the target, CRS OUT = the course from the target to the next waypoint (FCOM P 131). Type another one if ATC asks (NNN magnetic, NNNT true).

4. Check the temporary flight plan (TMPY, yellow) and press **INSERT DIR TO**.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/DIRECT-TO

DIRECT TO

UTC

DIST

OPTIONS

DIRECT

DIRECT WITH ABEAM

CRS IN

CRS OUT

RETURN

MSG LIST

Figure 2. The DIRECT TO page: target, UTC and distance, and the four OPTIONS.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/DIRECT-TO

DIRECT TO

UTC

DIST

OPTIONS

DIRECT

DIRECT WITH ABEAM

CRS IN

CRS OUT

Figure 3. CRS IN selected: the course field (amber boxes until a course is known).

With DIRECT WITH ABEAM, the temporary F-PLN shows **(ABEAM PTS)** in place of the abeam points until the direct is inserted (FCOM P 162).

3. What to expect

Situation	What you should see
DIRECT WITH ABEAM, waypoints bypassed	One ABxxxxx point per bypassed waypoint, on the direct leg.
DIRECT WITH ABEAM, a waypoint whose abeam is not between the aircraft and the target	No abeam point for it.
CRS IN / OUT with a course that can be intercepted ahead	INTCPT on the course, then the course.
CRS IN more than 160° from the current track, parallel, or behind	No intercept point: a course to the target.
CRS OUT parallel to the track	No intercept point: a course from the target.
CRS IN / OUT without a course	No DIR TO until a course is entered.

4. Limitations and why

Limitation	Why	What to do
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Abeam point names can repeat letters of the waypoint.	The FCOM rule is AB + 5 characters; two waypoints starting with the same 5 letters give the same name.	-
No intercept point when the geometry does not allow one.	Following the FCOM limits (160°, parallel, behind).	Use DIRECT, or another course.